ML Systems 101: From Jupyter to Production

Scaler School of Technology | ML System Design and LLMOps | Week 1 — Session 1/2 | Pre-Lecture Prep

What Is This Lecture About?

This session bridges everything you have learned in previous courses with production reality. You have trained models, evaluated them, maybe deployed one. Today you will learn what it actually takes for a model to serve millions of users reliably — and why most notebook pipelines fall apart the moment they hit the real world.

The Notebook-to-Production Gap

Why a model that works on your laptop is not a product. The six components every production ML system needs: data collection, feature store, training pipeline, model registry, serving layer, and monitoring.

Failure Modes in Production

Training-serving skew, silent degradation, data drift, feedback loops, and technical debt. Why models fail silently — and how to stop it.

The ML System Design Framework

A five-question framework (Problem → Data → Model → Serving → Monitoring) you will apply to every system we design in this course.

What to Expect in the Session

- Opening discussion: how would you deploy your best model right now?
- The six-component system anatomy diagram
- Two short quizzes on failure modes
- Sculley et al. (2015) technical debt discussion
- Live tooling demo: the gap between a notebook and a real pipeline
- Course roadmap: what the next 8 weeks look like

How to Prepare

- Think back to any model you have deployed or tried to deploy — what did you have to figure out yourself?
- Recall your feature engineering process from NLP — where did those features live at serving time?
- Optional: skim the abstract of Sculley et al. 2015 "Hidden Technical Debt in ML Systems" (Google it — free on NeurIPS)
- Ensure you have Google Colab access with a standard runtime
- No new tools needed today — we are setting the mental model first

Duration: ∼2 hours **Prerequisites:** NLP & Transformers, Applied AI and ML **Instructor:** Priyansh Saxena